

CREATE:

|                                                     |                             |                                               |                                                                                                                                                                            |                                                |
|-----------------------------------------------------|-----------------------------|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| Abstract—<br>loop    Reward<br>Evidence             | Editing    ”<br>Agent    ”  | CREATE<br>with    Training<br>Gymnasium<br>L1 | Closed-<br>CREATE                                                                                                                                                          | TABLE I<br>CREATE                              |
| CREATE<br>L2<br>Index   Terms—                      |                             |                                               | L1   L2   L3                                                                                                                                                               |                                                |
|                                                     | I.                          |                                               |                                                                                                                                                                            |                                                |
|                                                     | RL<br>Text2Reward<br>”    ” | Eureka                                        | episode<br>$J_t = \frac{1}{N} \sum_{i=1}^N \sum_{k=0}^{H_i-1} r_{\text{env}}(s_k^i, a_k^i, s_{k+1}^i).$                                                                    | (3)                                            |
|                                                     | CREATE                      | CREATE                                        | $B \quad n \quad C(n) = nB$                                                                                                                                                |                                                |
| — —                                                 |                             | ”    ”<br>B.                                  | —                                                                                                                                                                          |                                                |
|                                                     | episode                     | $R_t \rightarrow R_{t+1}$                     | $R_t = \sum_{j=1}^{m_t} r_{t,j}$                                                                                                                                           | CREATE                                         |
| A.                                                  | II.                         |                                               | $\alpha_{t,j} = \frac{\sum_k \mathbb{I}( r_{t,j}^{(k)}  > \epsilon)}{L_t}, \quad \rho_{t,j} = \frac{\sum_k  r_{t,j}^{(k)} }{\sum_{\ell,k}  r_{t,\ell}^{(k)}  + \epsilon}.$ | (4)                                            |
|                                                     |                             |                                               | CREATE<br>$E_t = (T_t, D_t, M_t)$                                                                                                                                          | episode                                        |
| B.                                                  |                             | C.                                            |                                                                                                                                                                            |                                                |
| Language   to<br>Eureka                             | Rewards                     | Text2Reward<br>CARD<br>RF-Agent<br>CREATE     | $E_t \quad h_t \quad j_t^* \quad z_t$<br>L1   L2                                                                                                                           | L1   L2   L3                                   |
| REvolve                                             |                             |                                               | $ \mathcal{K}_t^{\text{edit}}  = 1,$                                                                                                                                       | $R_{t+1} = \mathcal{U}(R_t, h_t, j_t^*, z_t).$ |
|                                                     |                             |                                               |                                                                                                                                                                            | (5)                                            |
| C.                                                  |                             |                                               | $\alpha_{t,j} \quad = \quad 0$                                                                                                                                             | episode                                        |
| ReAct                                               |                             | — —   Reflexion<br>CREATE                     | D.                                                                                                                                                                         |                                                |
|                                                     | Voyager                     |                                               | $D_{t+1} \quad T_{t+1}$                                                                                                                                                    |                                                |
| D.                                                  |                             |                                               | RL                                                                                                                                                                         | $M_t \quad A_t^*$                              |
|                                                     | CREATE                      |                                               |                                                                                                                                                                            |                                                |
|                                                     |                             | L1/L2/L3                                      |                                                                                                                                                                            |                                                |
|                                                     | III. CREATE                 |                                               | IV.                                                                                                                                                                        |                                                |
| A.                                                  |                             | A.                                            | Gymnasium   LunarLander-v3   BipedalWalker-v3<br>Stable-Baselines3   PPO   0—                                                                                              |                                                |
| CREATE                                              | I                           | $t \quad 4$                                   | II                                                                                                                                                                         |                                                |
|                                                     |                             |                                               | TABLE II                                                                                                                                                                   |                                                |
|                                                     | $x_t = (R_t, M_t, A_t^*),$  | (1)                                           |                                                                                                                                                                            |                                                |
| $R_t \quad M_t \quad A_t^* \quad -$                 |                             |                                               | /                                                                                                                                                                          |                                                |
| $(\pi_t, T_t) = \text{Train}(\mathcal{E}, R_t; B),$ | (2)                         |                                               | LunarLander   1M   20/100   200   5<br>BipedalWalker   5M   20/100   300   5                                                                                               |                                                |

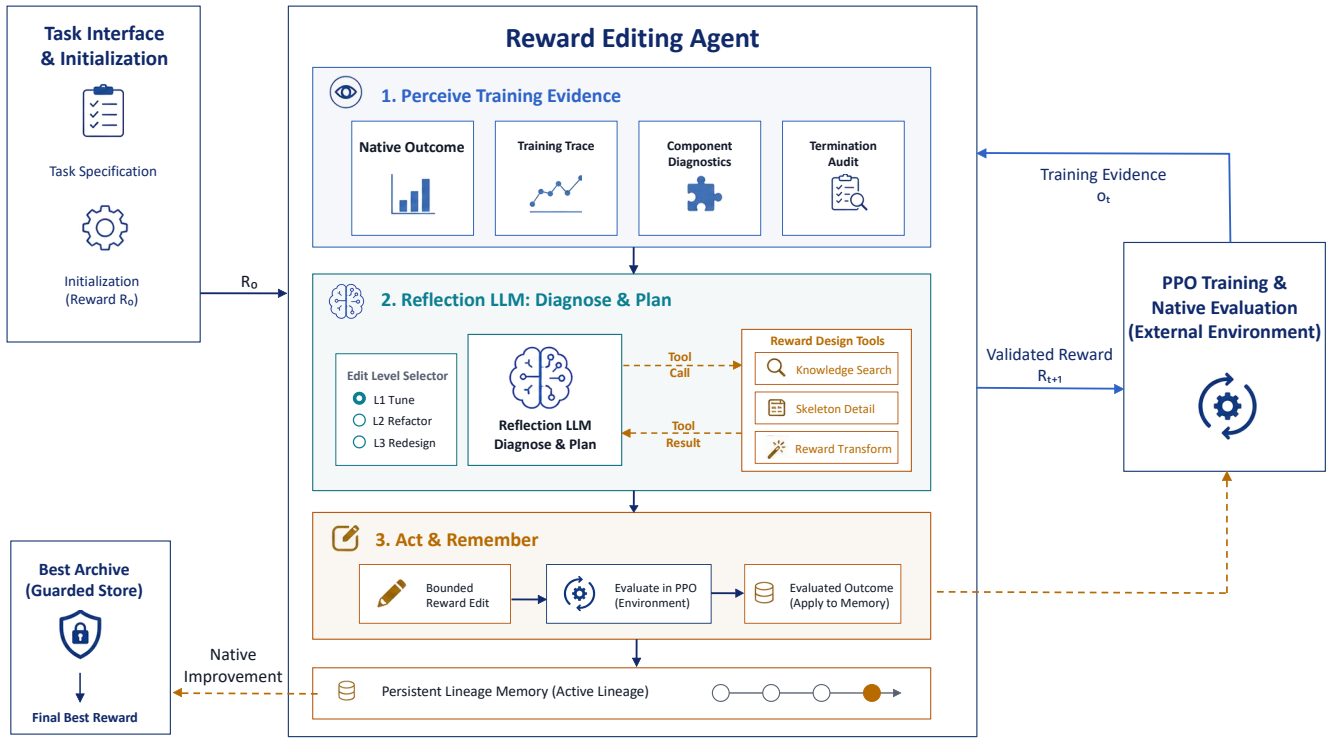


Fig. 1. CREATE  $R_0$  PPO  $J_t$  L1/L2/L3  $R_{t+1}$

20 episode PPO CREATE  $[-20, 20]$  100  
L1/L2/L3

TABLE III  
LunarLander

|        |   |   |     |               |
|--------|---|---|-----|---------------|
|        | - | - | 0/5 | -2.21         |
|        | - |   | 0/5 | -0.74         |
|        |   |   | 2/5 | 134.97        |
|        |   |   | 0/5 | 114.21        |
| CREATE |   |   | 5/5 | <b>228.98</b> |

CREATE

III

8 10 2 4 3 27

B.

2

LunarLander

CREATE  
episode

120.02 L2

L1

-29.48  
-17.90

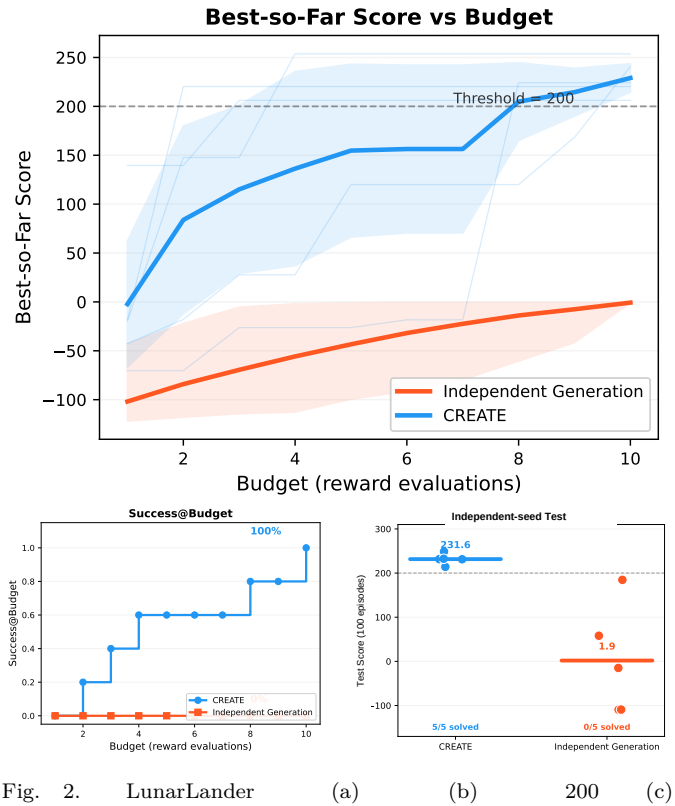


Fig. 2. LunarLander (a) (b) 200 (c)